

量子力学Ⅱ 第4-6回 確認問題 解答例

1.

$$\begin{aligned}\hat{J}_- \hat{J}_+ &= (\hat{J}_1 - i\hat{J}_2)(\hat{J}_1 + i\hat{J}_2) \\ &= \hat{J}_1^2 + \hat{J}_2^2 + i(\hat{J}_1 \hat{J}_2 - \hat{J}_2 \hat{J}_1) \\ &= \hat{\mathbf{J}}^2 - \hat{J}_3^2 - \hbar \hat{J}_3\end{aligned}$$

より

$$\begin{aligned}\langle j, m | \hat{J}_- \hat{J}_+ | j, m \rangle &= \langle j, m | (\hat{\mathbf{J}}^2 - \hat{J}_3^2 - \hbar \hat{J}_3) | j, m \rangle \\ c_{jm}^* \langle j, m+1 | \cdot c_{jm} | j, m+1 \rangle &= \{j(j+1) - m(m+1)\} \hbar^2 \langle j, m | j, m \rangle\end{aligned}$$

$$c_{jm} = \sqrt{j(j+1) - m(m+1)} \hbar$$

$$\begin{aligned}\hat{J}_+ \hat{J}_- &= (\hat{J}_1 + i\hat{J}_2)(\hat{J}_1 - i\hat{J}_2) \\ &= \hat{J}_1^2 + \hat{J}_2^2 - i(\hat{J}_1 \hat{J}_2 - \hat{J}_2 \hat{J}_1) \\ &= \hat{\mathbf{J}}^2 - \hat{J}_3^2 + \hbar \hat{J}_3\end{aligned}$$

より

$$\begin{aligned}\langle j, m | \hat{J}_+ \hat{J}_- | j, m \rangle &= \langle j, m | (\hat{\mathbf{J}}^2 - \hat{J}_3^2 + \hbar \hat{J}_3) | j, m \rangle \\ d_{jm}^* \langle j, m-1 | \cdot d_{jm} | j, m-1 \rangle &= \{j(j+1) - m(m-1)\} \hbar^2 \langle j, m | j, m \rangle\end{aligned}$$

$$d_{jm} = \sqrt{j(j+1) - m(m-1)} \hbar$$

2.

(1)

$$Y_0^0(\theta, \varphi) = \frac{1}{\sqrt{4\pi}}$$

$$Y_1^{\pm 1}(\theta, \varphi) = \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\varphi}, \quad Y_1^0(\theta, \varphi) = \sqrt{\frac{3}{4\pi}} \cos \theta$$

(2)

$$\begin{aligned}\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) \sin \theta e^{\pm i\varphi} &= e^{\pm i\varphi} \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \cos \theta) \\ &= e^{\pm i\varphi} \frac{1}{\sin \theta} (\cos^2 \theta - \sin^2 \theta) = e^{\pm i\varphi} \frac{1}{\sin \theta} (1 - 2 \sin^2 \theta)\end{aligned}$$

$$\frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \sin \theta e^{\pm i\varphi} = \frac{1}{\sin^2 \theta} \sin \theta \frac{\partial^2}{\partial \varphi^2} e^{\pm i\varphi} = -\frac{1}{\sin \theta} e^{\pm i\varphi}$$

より

$$\hat{\mathbf{L}}^2 Y_1^{\pm 1}(\theta, \varphi) = 2\hbar^2 Y_1^{\pm 1}(\theta, \varphi)$$

$$\begin{aligned}
\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) \cos \theta &= \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (-\sin^2 \theta) \\
&= \frac{1}{\sin \theta} (-2 \sin \theta \cos \theta) = -2 \cos \theta \\
\frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \cos \theta &= 0
\end{aligned}$$

より

$$\hat{\mathbf{L}}^2 Y_1^0(\theta, \varphi) = 2\hbar^2 Y_1^0(\theta, \varphi)$$

また,

$$-i\hbar \frac{\partial}{\partial \varphi} \sin \theta e^{\pm i\varphi} = \pm \hbar \sin \theta e^{\pm i\varphi}, \quad -i\hbar \frac{\partial}{\partial \varphi} \cos \theta = 0$$

より

$$\hat{L}_z Y_1^{\pm 1} = \pm \hbar Y_1^{\pm 1}, \quad \hat{L}_z Y_1^0 = 0$$

以上より, $Y_1^{\pm 1}, Y_1^0$ は $\hat{\mathbf{L}}^2$ の固有関数で固有値は $2\hbar^2$. また, $Y_1^{\pm 1}, Y_1^0$ は $\hat{L}_z$ の固有関数で固有値は $\pm \hbar, 0$.

Y_0^0 は定数だから, $\hat{\mathbf{L}}^2 Y_0^0 = 0, \hat{L}_z Y_0^0 = 0$. よって, Y_0^0 は $\hat{\mathbf{L}}^2, \hat{L}_z$ の固有関数で, 固有値はともに 0.

3.

(1)

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} R \right) + \frac{l(l+1)\hbar^2}{2mr^2} R - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r} R = ER$$

において $r = r_0 \rho, E = -E_0 \epsilon$ とおくと

$$\begin{aligned}
-\frac{\hbar^2}{2m} \frac{1}{r_0^2} \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial}{\partial \rho} \tilde{R} \right) + \frac{l(l+1)\hbar^2}{2mr_0^2} \frac{1}{\rho^2} \tilde{R} - \frac{e^2}{4\pi\epsilon_0} \frac{1}{r_0} \frac{1}{\rho} \tilde{R} &= -E_0 \epsilon \tilde{R} \\
\frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial}{\partial \rho} \tilde{R} \right) - \frac{l(l+1)}{\rho^2} \tilde{R} + \frac{e^2}{4\pi\epsilon_0} \frac{2m}{\hbar^2} r_0 \frac{1}{\rho} \tilde{R} &= \frac{2mE_0}{\hbar^2} r_0^2 \epsilon \tilde{R}
\end{aligned}$$

よって,

$$\frac{e^2}{4\pi\epsilon_0} \frac{2m}{\hbar^2} r_0 = 2, \quad \frac{2mE_0}{\hbar^2} r_0^2 = 1$$

とおけばよい. これより

$$r_0 = \frac{4\pi\epsilon_0 \hbar^2}{me^2}$$

$$E_0 = \frac{\hbar^2}{2mr_0^2} = \frac{\hbar^2}{2m} \left(\frac{me^2}{4\pi\epsilon_0\hbar^2} \right)^2 = \frac{1}{2} \left(\frac{e^2}{4\pi\epsilon_0\hbar c} \right)^2 mc^2$$

(2) $l = 0$ の場合の (1) の方程式に $\tilde{R} = e^{-\rho}$ を代入すると

$$\frac{d^2}{d\rho^2}e^{-\rho} + \frac{2}{\rho}\frac{d}{d\rho}e^{-\rho} + \frac{2}{\rho}e^{-\rho} = \epsilon e^{-\rho}$$

$$e^{-\rho} = \epsilon e^{-\rho}$$

よって, $\epsilon = 1$

(3) $l = 0$ の場合の (1) の方程式に $\tilde{R} = (1 + a\rho)e^{-\rho/2}$ を代入すると

$$\frac{d^2}{d\rho^2}(1 + a\rho)e^{-\rho/2} + \frac{2}{\rho}\frac{d}{d\rho}(1 + a\rho)e^{-\rho/2} + \frac{2}{\rho}(1 + a\rho)e^{-\rho/2} = \epsilon(1 + a\rho)e^{-\rho/2}$$

$$\left(-a + \frac{1}{4} + \frac{1}{4}a\rho\right)e^{-\rho/2} + \frac{2}{\rho}\left(a - \frac{1}{2} - \frac{1}{2}a\rho\right)e^{-\rho/2} + \frac{2}{\rho}(1 + a\rho)e^{-\rho/2} = \epsilon(1 + a\rho)e^{-\rho/2}$$

$$\left(\frac{1 + a\rho}{4} + \frac{2a + 1}{\rho}\right)e^{-\rho/2} = \epsilon(1 + a\rho)e^{-\rho/2}$$

よって

$$a = -\frac{1}{2}, \quad \epsilon = \frac{1}{4}$$

(4) $l = 1$ の場合の (1) の方程式に $\tilde{R} = \rho e^{-\rho/2}$ を代入すると

$$\frac{d^2}{d\rho^2}\rho e^{-\rho/2} + \frac{2}{\rho}\frac{d}{d\rho}\rho e^{-\rho/2} - \frac{2}{\rho^2}\rho e^{-\rho/2} + \frac{2}{\rho}\rho e^{-\rho/2} = \epsilon\rho e^{-\rho/2}$$

$$\left(-1 + \frac{1}{4}\rho\right)e^{-\rho/2} + \frac{2}{\rho}\left(1 - \frac{1}{2}\rho\right)e^{-\rho/2} - \frac{2}{\rho^2}\rho e^{-\rho/2} + \frac{2}{\rho}\rho e^{-\rho/2} = \epsilon\rho e^{-\rho/2}$$

$$\frac{1}{4}\rho e^{-\rho/2} = \epsilon\rho e^{-\rho/2}$$

よって

$$\epsilon = \frac{1}{4}$$

(5)

$$\tilde{R}(\rho) = Ce^{-\rho}$$

規格化条件は

$$\int_0^\infty R(r)^2 r^2 dr = 1$$

より

$$r_0^3 \int_0^\infty \tilde{R}(\rho)^2 \rho^2 d\rho = 1$$

$$r_0^3 |C|^2 \int_0^\infty \rho^2 e^{-2\rho} d\rho = 1$$

$$\int_0^\infty \rho^n e^{-\alpha\rho} d\rho = \frac{n!}{\alpha^{n+1}}$$

より

$$r_0^3 |C|^2 \frac{2!}{2^3} = 1, \quad C = \sqrt{\frac{4}{r_0^3}}$$

$$\phi_{1s}(\boldsymbol{r}) = \tilde{R}(\rho) Y_0^0 = \sqrt{\frac{1}{\pi r_0^3}} e^{-\frac{r}{r_0}}$$

$$\langle r \rangle = \frac{\int_0^\infty r e^{-\frac{2r}{r_0}} r^2 dr}{\int_0^\infty e^{-\frac{2r}{r_0}} r^2 dr}$$

$$= r_0 \frac{\int_0^\infty \rho^3 e^{-2\rho} d\rho}{\int_0^\infty \rho^2 e^{-2\rho} d\rho}$$

$$\langle r \rangle = r_0 \frac{\frac{3!}{2^4}}{\frac{2!}{2^3}} = \frac{3}{2} r_0$$